

Solution Set: Gravity Anomalies and Isostatic Response

Question 2: Shallow and Deep Bodies

(A) Gravity anomaly of a 2-D polygon

The Talwani et al. (1959) formulation gives the exact vertical gravity anomaly for a two-dimensional body of constant density contrast $\Delta\rho$ with polygonal geometry. The method assumes:

- infinite extent perpendicular to the x - z plane,
- constant density contrast,
- observation at the surface ($z = 0$).

The total anomaly is obtained as the sum of contributions from each polygon edge. No approximation is made beyond the 2-D assumption. Implementation is most easily performed by looping over polygon edges in Python or MATLAB.

(B) Triangular rift basin

(1) Geometry

The basin is bounded by two faults dipping at 60° and has a total vertical offset of 5 km. The basin is centered at $(0, 0)$, with the surface at $z = 0$.

The horizontal half-width of the basin is:

$$w = \frac{(5/2)}{\tan 60^\circ} \approx 1.44 \text{ km}.$$

A suitable clockwise ordering of vertices is:

$$(-1.44, 0), \quad (0, -5), \quad (1.44, 0).$$

(2) Gravity anomaly

Using the polygon equation with a negative density contrast typical of sediments ($\Delta\rho \approx -400 \text{ kg m}^{-3}$), the computed anomaly is negative and symmetric. The anomaly decays smoothly to zero at large distances from the basin.

(3) Interpretation

- The maximum horizontal gradient occurs near the basin edges, not at the basin center.
- Because the geometry is symmetric, the anomaly is symmetric about $x = 0$.

The gravity signal is dominated by the sloping fault planes rather than the deepest point of the basin.

(C) Trapezoidal rift basin

(1) Geometry

The trapezoidal basin has the same surface coordinates as part (B) but a flat base at 3 km depth. The sidewall dips are comparable to those in the triangular case. One valid vertex ordering is:

$$(-1.44, 0), (-0.44, -3), (0.44, -3), (1.44, 0).$$

(2) Gravity anomaly

The trapezoidal basin produces a broader, smoother anomaly compared to the triangular basin, with reduced curvature near the edges.

(3) Comparison

- **Edge slope** controls the location and sharpness of gradients.
- **Depth extent** controls anomaly amplitude and wavelength.
- **Density contrast** scales the magnitude but does not strongly affect shape.

Different basin geometries with similar edges and depths can produce very similar gravity anomalies.

(D) Basin over a crustal root

The crustal root is represented by a triangular body of width 30 km and maximum thickness 10 km. Typical density contrasts are:

$$\Delta\rho_{\text{basin}} \sim -400 \text{ kg m}^{-3}, \quad \Delta\rho_{\text{root}} \sim +300 \text{ kg m}^{-3}.$$

Root alone

The root produces a long-wavelength, low-amplitude anomaly dominated by its depth and width.

Basin alone

The basin produces a short-wavelength anomaly dominated by edge effects.

Combined model

When the basin and root are combined, the shallow basin dominates the short-wavelength signal, while the root affects the long-wavelength background. With suitable basin parameters, the combined anomaly can closely resemble a basin-only anomaly.

This demonstrates the fundamental non-uniqueness of gravity interpretation.

Question 3: Isostasy — Oceanic vs Continental Lithosphere

Isostatic columns constructed using crustal thickness alone cannot balance oceanic and continental lithosphere. Balance requires a density difference within the mantle lithosphere of order:

$$\Delta\rho \sim 30\text{--}60 \text{ kg m}^{-3},$$

which can plausibly be explained by compositional or thermal differences.

Question 4: Isostatic Response to Erosion

(A) Single-layer crust

Assume:

$$\rho_c = 2800 \text{ kg m}^{-3}, \quad \rho_m = 3300 \text{ kg m}^{-3}.$$

The Airy isostatic factor is:

$$\alpha = \frac{\rho_m - \rho_c}{\rho_m} \approx 0.15.$$

(1) Instantaneous erosion of 2 km

Surface lowering:

$$\Delta \varepsilon = 2 \text{ km} \times \alpha \approx 0.30 \text{ km}.$$

New elevation:

$$\varepsilon \approx 5.8 \text{ km}.$$

(2) Total erosion to baseline

$$E = \frac{\varepsilon_0}{\alpha} \approx \frac{6.1}{0.15} \approx 41 \text{ km}.$$

(3) Fraction replaced by rebound

$$1 - \alpha \approx 85\%.$$

(B) Continuous erosion with rebound

The erosion law is:

$$\frac{dE}{dt} = k\varepsilon, \quad k = 0.1 \text{ Myr}^{-1}.$$

Under Airy isostasy:

$$\frac{d\varepsilon}{dt} = -\alpha \frac{dE}{dt} = -\alpha k\varepsilon.$$

(1) General solution

$$\varepsilon(t) = \varepsilon_0 e^{-\alpha k t}.$$

(2) No rebound limit

If $\alpha = 1$:

$$\varepsilon(t) = \varepsilon_0 e^{-k t}.$$

(3) Initial erosion rates

With $\varepsilon_0 = 6100 \text{ m}$:

$$\left. \frac{dE}{dt} \right|_{t=0} = 610 \text{ m Myr}^{-1}, \quad \left. \frac{d\varepsilon}{dt} \right|_{t=0} \approx 91 \text{ m Myr}^{-1}.$$

(4) Time to erode 95% of relief

$$t_{95} = \frac{\ln(20)}{\alpha k} \approx 200 \text{ Myr}.$$

(5) Interpretation

Isostatic rebound increases erosion timescales by more than an order of magnitude, allowing large mountain belts to persist for hundreds of millions of years despite vigorous erosion.